Independent Project Introduction

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```
# load data
arthropod_ranch <- read.csv("data/sample_taxa_matrix_by_ranch_WEIGHTED_REL.csv")
arthropod_trap  <- read.csv("data/sample_taxa_matrix_by_trap_WEIGHTED_REL.csv")
salience        <- read.csv("data/3_1salience_plants_personal_use.csv")
```

Introduction

Arid and fog-influenced desert systems are characterized by strong climatic variability, where episodic rainfall and ENSO (El Niño Southern Oscillation) dynamics shape the patterns of productivity, species interactions, and long-term ecosystem stability (Ardoin et al., 2022; Grismer, 2002). In Baja California Sur, Mexico, the community of San Juanico lies within a transition zone between desert, coastal lagoon and oasis systems, creating a biologically diverse landscape shaped by environmental gradients such as elevation (Stiner et al., 2025). Despite documented climatic stress in the region, fine-scale monitoring efforts often rely on short-term or externally driven sampling, leaving gaps in understanding how community-level metrics vary across space. (Danielsen et al., 2009; Reed, 2008).

Arthropods are widely used as ecological indicators because their assemblages shift predictably across environmental gradients and seasonal conditions, allowing measurable changes in abundance and diversity to reflect broader ecological patterns (Doran et al., 2003). The ranches surrounding San Juanico span a clear elevational gradient, providing an opportunity to evaluate how insect communities vary across space. Using malaise trap collections from multiple ranch sites, this project quantifies arthropod richness at the trap level and tests whether richness varies along an elevational gradient. Establishing this baseline relationship provides a simple measure of spatial variation in arthropod communities in a desert system experiencing environmental

(1) Ardoin, N. M., et al. 2022. Ecosystems & People 18. (2) Journal of Nature Conservation 22. (3) EO Stiner et al. 2025. Thematic Analysis of Climatological Challenges in Baja Sur. (4) Danielsen, F., et al. 2009. Conservation Biology 23. (5) Doran, N. 2003. Organisms Diversity & Evolution 3(2). (6) Grismer, L. L. 2002. University of California Press. (7) Reed, M. S. 2008. Biological Conservation 141(10).

Research Question

Does insect richness vary along an elevational gradient across ranches in a San Juanico Baja California system?

Hypotheses

H0: Insect richness does not vary with elevation.
HA: Insect richenss varies with elevation.

Aim 1:

Quantify insect richness at the trap level and test whether richness varies along an elevational gradient across ranches using linear regression.

Scope of the project

For this course project, I focus on a simplified analysis of insect community variation across ranch sites in the San Juanico system. Using malaise trap data, iinsect richness was quantified at the trap level and analyzed using a regression analysis in relation to elevation as an environmental gradient. The analysis is limited to testing whether richness varies with elevation using a linear regression approach.

This is a simplified component of a broader research project that includes additional environmental variables, plant community data, and participatory research elements, which were not included in this assignment.

Statistical Approach

Insect richness was calculated at the trap level by counting the number of taxa with nonzero abundance values from a weighted abundance matrix. Elevation values for each trap were obtained using Google Earth based on mapped trap locations.

A linear regression was used to test whether or not arthropod richness varies with elevation across traps. Elevation was chosen as the continuous predictor and richess was chosen as the response variable. A scatterplot with a fitted regression line was used to visualize the relationship between elevation and richness.

```
abund <- read.csv("data/sample_taxa_matrix_by_trap_WEIGHTED_REL.csv")
names(abund)
```

```
##  [1] "taxon_key"           "taxon_key.1"         "genus_origin"
##  [4] "genus"               "phylum"              "class"
##  [7] "order"               "family"              "species"
## [10] "n_species_collapsed" "Casa_Vieja_TE"       "Casa_Vieja_UT"
## [13] "El_Mezquital_F"      "El_Mezquital_H"      "El_Tule_BH"
## [16] "El_Tule_UT"          "La_Venada_1"         "La_Venada_2"
## [19] "NC_2_S124"
```

```
# select only trap columns
trap_data <- abund[, c(
  "Casa_Vieja_TE", "Casa_Vieja_UT",
  "El_Mezquital_F", "El_Mezquital_H",
  "El_Tule_BH", "El_Tule_UT",
  "La_Venada_1", "La_Venada_2"
)]

# calculate richness (count values > 0 per column)
richness <- colSums(trap_data > 0)

richness
```

```
##  Casa_Vieja_TE  Casa_Vieja_UT El_Mezquital_F El_Mezquital_H     El_Tule_BH
##             33             33             29             44             12
##     El_Tule_UT    La_Venada_1    La_Venada_2
##             57             47             59
```

```r
data <- read.csv("data/richness_elevation.csv")

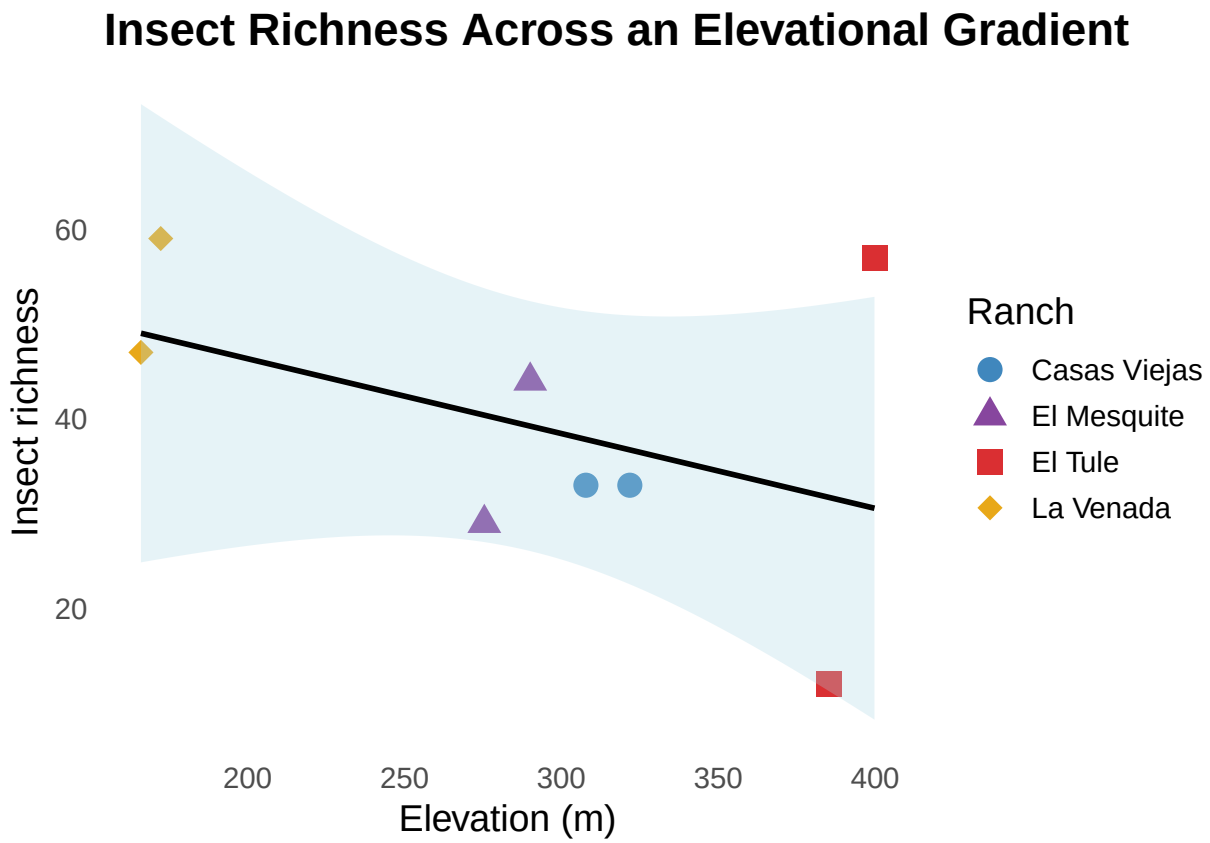
head(data)
```

```
##          Ranch Trap Latitude Longitude Elevation_m Richness
## 1    La Venada    1 26.49627 -112.4615      165.97       47
## 2    La Venada    2 26.49612 -112.4633      172.28       59
## 3  El Mesquite    1 26.56285 -112.4006      275.42       29
## 4  El Mesquite    2 26.56494 -112.3996      290.05       44
## 5 Casas Viejas    1 26.62410 -112.3573      307.80       33
## 6 Casas Viejas    2 26.62564 -112.3551      321.80       33
```

```r
library(ggplot2)

ggplot(data, aes(x = Elevation_m, y = Richness, color = Ranch, shape = Ranch)) +
  geom_point(size = 4, alpha = 0.9) +
  geom_smooth(
    data = data,
    aes(x = Elevation_m, y = Richness),
    method = "lm",
    se = TRUE,
    fill = "lightblue",
    color = "black",
    alpha = 0.3,
    linewidth = 1,
    inherit.aes = FALSE
  ) +
  labs(
    title = "Insect Richness Across an Elevational Gradient",
    x = "Elevation (m)",
    y = "Insect richness",
    color = "Ranch",
    shape = "Ranch"
  ) +
  scale_color_manual(values = c(
    "Casas Viejas" = "#2C7BB6",
    "El Mesquite"  = "#7B3294",
    "El Tule"      = "#D7191C",
    "La Venada"    = "#E69F00"
  )) +
  scale_shape_manual(values = c(
    "Casas Viejas" = 16,
    "El Mesquite"  = 17,
    "El Tule"      = 15,
    "La Venada"    = 18
  )) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "right",
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank()
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
model <- lm(Richness ~ Elevation_m, data = data)
summary(model)
```

```
##
## Call:
## lm(formula = Richness ~ Elevation_m, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -19.713  -6.469  -2.873   6.197  26.426
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 62.11086   20.09376   3.091   0.0214 *
## Elevation_m -0.07888    0.06680  -1.181   0.2823
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.22 on 6 degrees of freedom
## Multiple R-squared:  0.1886, Adjusted R-squared:  0.05334
## F-statistic: 1.394 on 1 and 6 DF,  p-value: 0.2823
```

Results

Arthopod richness showed a negative relationship with elevation (slope = -0.079); however, this relationship was not statistically significant (p=0.282). Elevation accounted for only a small proportion of the variation in richness R2= 0.19). A slight decreasing trend in richness with increasing elevation was noticed, variability among traps remained high.